Project Overview

Customer Churn Analysis & Prediction System

End-to-End ETL, Business Intelligence & Machine Learning Solution

- SQL Server based ETL pipeline
- Power BI Executive Dashboard
- Random Forest ML Model for Churn Prediction
- Business insights for retention strategy

Objective: To analyze historical churn patterns and predict future churners using data-driven decision making.

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Introduction to Churn Analysis

What is Churn?

Churn refers to customers who stop doing business with a company.

In telecom:

- Subscription cancellation
- Non-renewal of contract
- Switching to competitor

Why Churn Matters

- Acquiring new customers costs 5x more than retaining existing ones
- High churn directly reduces revenue
- Understanding churn improves marketing & retention strategy

Goal: Identify why customers leave and predict who might leave next.

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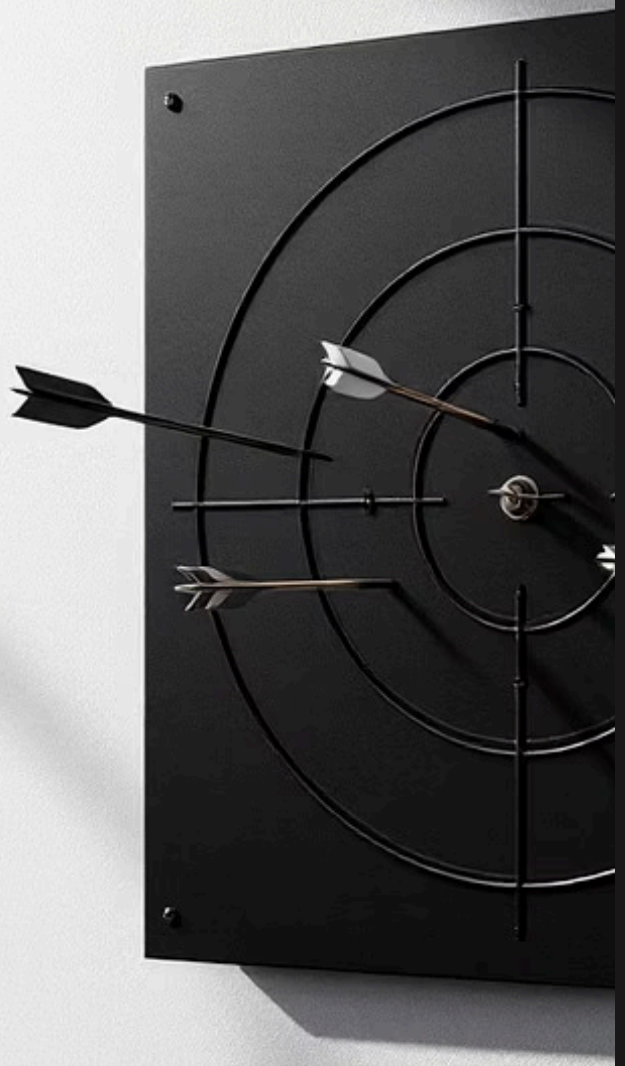

Business Objectives

Project Targets

1. Visualize and analyze customer data across:
    - Demographics
    - Geographic distribution
    - Payment & account information
    - Services used
2. Study churner profile
3. Identify marketing opportunities
4. Predict future churners using Machine Learning

Key Metrics

- Total Customers
- Total Churn
- Churn Rate
- New Joiners

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Data Architecture & ETL Process

Step 1 – SQL Server ETL

- Created database: db_Churn
- Imported CSV into staging table
- Cleaned null values
- Standardized categorical values
- Inserted cleaned data into production table
- Created views:
  - vw_ChurnData
  - vw_JoinData

Why SQL?

- Better data integrity
- Scalable system
- Industry-standard database management
- Supports recurring data loads

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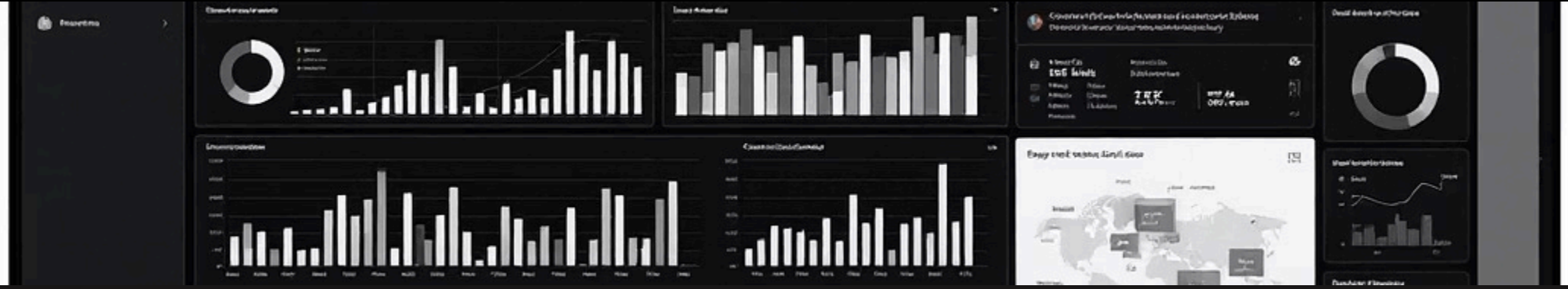

Data Transformation in Power BI

Calculated Columns Created

- Churn Status (1 = Churned, 0 = Stayed)
- Monthly Charge Range
- Age Groups
- Tenure Groups

Data Modeling Enhancements

- Created mapping tables
- Unpivoted services data
- Built relationships
- Created DAX measures:
  - Total Customers
  - Total Churn
  - Churn Rate
  - New Joiners

This enabled dynamic dashboard analytics.

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Executive Summary Dashboard Insights

Overall Metrics

- Total Customers: 6,418
- Churn Rate: 27%
- Total Churn: 1,732
- New Joiners: 411

Key Insights

- Month-to-month contracts show highest churn
- Fiber optic users have higher churn
- Jammu region historically showed high churn
- Premium Support reduces churn significantly
- Competitor is primary churn reason

Conclusion: Contract type and service quality are major churn drivers.

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Churn Distribution & Customer Behavior Insights

Demographic Analysis

- Higher churn observed in older age groups
- Slight variation across gender

Service-Based Insights

- Fiber customers show higher churn
- Customers with Premium Support are more loyal
- Paperless billing users show higher churn tendency

Payment Insights

- Credit card autopay users show lower historical churn
- Manual payment methods show higher churn

Key Takeaway: Long-term contracts + premium services improve retention.

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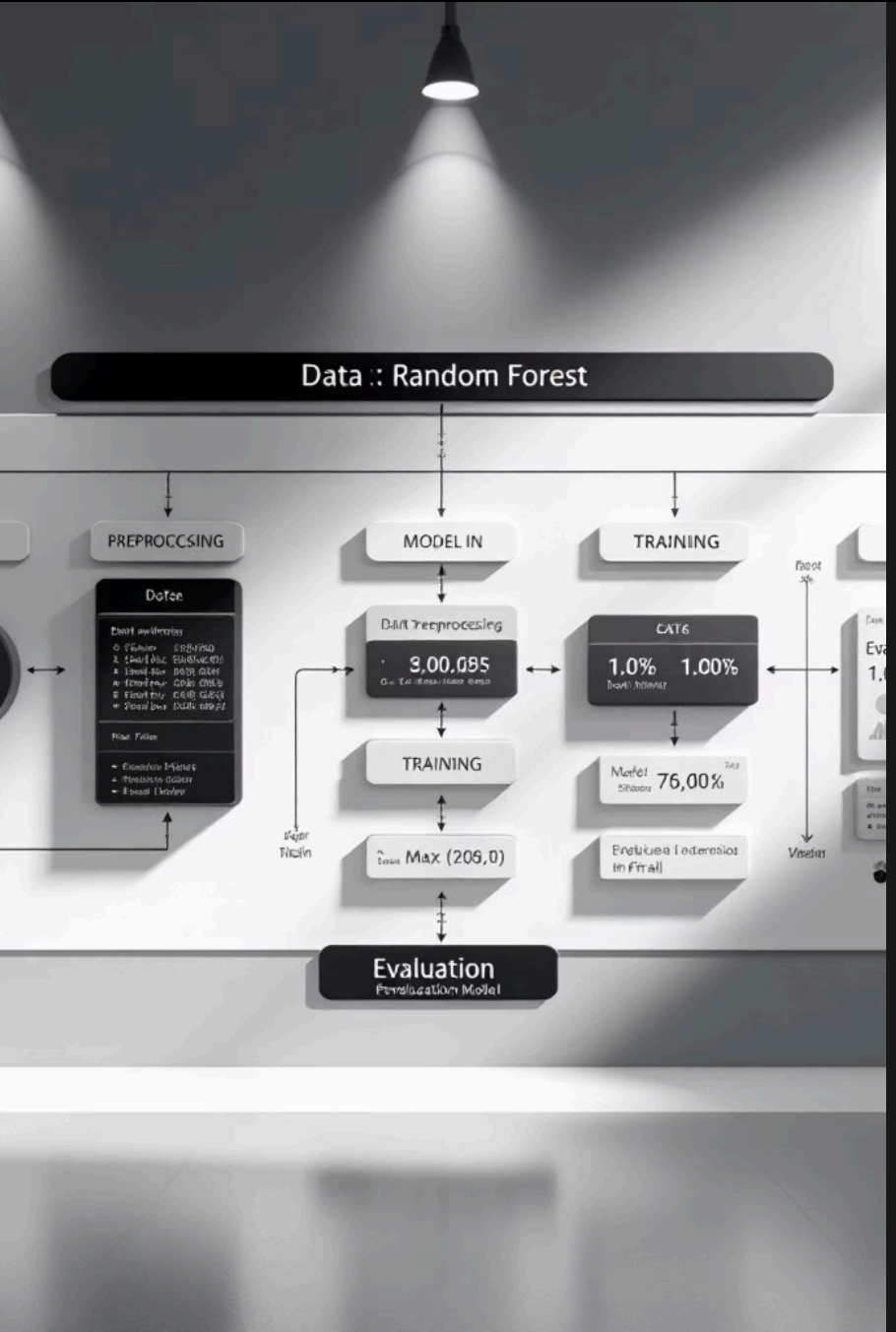

Machine Learning Model (Random Forest)

Why Random Forest?

- Handles categorical + numerical data well
- Reduces overfitting
- Strong classification performance

Model Process

- Label encoding categorical variables
- Train-test split (80-20)
- RandomForestClassifier (100 estimators)
- Evaluated using:
  - Confusion Matrix
  - Classification Report
  - Feature Importance

Outcome: Identified high-risk customers with strong predictive performance.

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Churn Prediction Results

Predicted Churners: 374 Customers

Prediction dashboard includes:

- Customer ID
- Monthly Charge
- Revenue
- Refunds
- Referrals
- Contract
- Payment Method
- State

Insight Difference

Historical churn ≠ Predicted churn

Prediction focuses on: Active customers with highest risk probability.

Enables proactive retention campaigns.

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Business Impact & Conclusion

Business Impact

- Identify high-risk customers early
- Enable targeted marketing campaigns
- Improve retention strategies
- Reduce revenue loss
- Data-driven decision making

Final Conclusion

This project successfully:

- Built an end-to-end ETL pipeline
- Created interactive business dashboard
- Developed predictive ML model
- Provided actionable churn insights

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